

AegisPulse: Real-time network intrusion detection using hybrid deep learning and live attack simulation

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Abstract: The attack surface, for actors grows as the network infrastructures become more complex. The conventional security solutions become inadequate. The static signature matching used by Network Intrusion Detection Systems (NIDS) makes the detection of assaults and zero-day exploits difficult. This study presents the AegisPulse framework. The AegisPulse framework combines a real-time assault simulation engine, with the Deep Learning (DL) architectures. On datasets such, as NSL-KDD and CICIDS2017 the system achieves accuracy. The system uses a combinations of Convolutional Neural Networks (CNN), for feature extraction & Deep Neural Networks (DNN) for classification. AegisPulse includes a live visualization dashboard. The dashboard solves the " box" issue of AI in cybersecurity. The interface gives security analysts information. The interface shows attack data in time.

Key Words: Network Intrusion Detection System (NIDS), Deep Learning, CNN, Real-Time Simulation, Cybersecurity, Traffic Analysis, Hybrid Models.

1. INTRODUCTION: The growth of cloud services and the Internet of Things (IoT) creates an amount of data exchange, in the era. The data flow is massive. I notice that the growth of cloud services and the Internet of Things (IoT) drives innovation. The growth of cloud services and the Internet of Things (IoT) also leaves infrastructure open to cyberattacks. DDoS attacks and ransomware attacks are cyberattacks that have grown from scripts, to automated campaigns that can bypass normal firewalls.

In my work I have seen that modern cybersecurity tactics must include the Network Intrusion Detection Systems (NIDS). New unknown attacks (zero-day threats) are invisible, to the signature-based systems. At the time the anomaly-based systems often have false alarms even though the anomaly-based systems can identify new dangers. Adding Artificial Intelligence (AI) into the Network Intrusion Detection Systems (NIDS) has become a focus to address these limits. In my work I see that Machine learning (ML) and Deep learning (DL) algorithms learn the features of network activity. I also see that machine learning (ML) and deep learning (DL) algorithms spot events that point to an attack.

Even though AI can do a lot the lack of interpretability still blocks its use, in work. I see security analysts often get classifications or raw logs that have no context. In this work I present AegisPulse, a NIDS solution. AegisPulse shows data in time. Also predicts threats using hybrid deep learning. AegisPulse works for both security and, for teaching. AegisPulse does not find threats, AegisPulse also. Shows those threats.

2. LITERATURE REVIEW: I have read that advanced neural computing now replaces techniques, in intrusion detection. Early research focused on anomaly detections that looked for departures from a baseline. The release of the KDD dataset marked a turning point because the dataset let researchers use data mining methods like Support Vector Machines (SVM). Researchers often need to do feature engineering for data mining methods such, as Support Vector Machines (SVM).

I have noticed that the better performance of Deep Learning (DL), in extracting features has been shown in publications. I have also seen that people have used Convolutional Neural Networks (CNNs) for image recognition. The Convolutional Neural Networks (CNNs) have also been changed to work for NIDS. I have read that the Convolutional Neural Networks (CNNs) can find relationships between packet attributes that the conventional ML models miss because the Convolutional Neural Networks (CNNs) treat network traffic flows as vectors.. Researchers test the majority of the suggested models, in batch processing and offline settings. The technical difficulty of putting the models into a streaming environment with a user facing dashboard is rarely covered in research. Researchers rarely discuss the difficulty. AegisPulse offers a system that's accurate, in backend processing and accessible in frontend presentation. The system is accurate in backend processing. The system is accessible, in frontend presentation. The system aims to close the gap. The technical difficulty remains a challenge. The models need to run. The dashboard needs to show the results. The research does not cover the difficulty. AegisPulse wants to close the gap.

3. OBJECTIVES / AIMS: The primary objectives of this research are:

- Create a Deep Learning model that sorts network traffic into benign categories. The hybrid Deep Learning model merges CNN and DNN architectures.
- I will create a Live Attack Simulation engine. The Live Attack Simulation engine will simulate real-time network performance by streaming dataset records.
- I will improve the NIDSs interpretability. I will build a web dashboard that shows the attack vectors and the geographic origins.

4. RESEARCH METHOD / METHODOLOGY: We built AegisPulse using a step-by-step process. We checked each step. AegisPulse used data collection, data cleaning, model design and full-stack system integration.

4.1 Data Acquisition & Preprocessing: In practice the system uses the NSL-KDD and CICIDS2017 datasets to handle both attacks and new attacks. The system developers built a preparation pipeline because raw network data cannot be fed directly into networks. To keep descent steady during training the system pipeline clears values applies One-Hot Encoding to characteristics (such, as protocols like TCP and UDP). Uses Min-Max Scaling to normalize numerical data (such, as packet time).

4.2 Hybrid Model Architecture: We built the hybrid model. The Convolutional Neural Network (CNN) branch looks at packet headers. Finds patterns by checking the data for small patterns, with 1D convolution. The Deep Neural Network (DNN) branch has layers. The Deep Neural Network (DNN) branch learns relationships between features. The Swallow Neural Network (SNN) is a network that can act as a backup, for response when speed matters.

4.3 Real-Time Simulation Engine: I built a Python replay engine that mimics a network without running viruses. The replay engine uses WebSocket to send processed rows from the test data sets to the dashboard. The replay engine can send at a rate you can set, for example 20 packets, per second. The replay engine lets the system be tested against streams by simulating the pulse of a network.

5. FINDINGS / RESULTS: I used performance indicators such, as accuracy, precision, recall and F1-score to assess the suggested AegisPulse system. The standard performance indicators gave a picture of how the AegisPulse system performed.

5.1 Classification Accuracy: In my experiments the hybrid CNN model achieved an accuracy of 95.2 percent, on the testing split of the CICIDS2017 dataset. The hybrid CNN model could clearly separate high-volume attacks such as DDoS from traffic. The DNN model reached an accuracy of 92 percent staying close, behind the CNN model.

5.2 Dashboard Performance: I observed that the visualization engine responded quickly. The dashboard rendered than 50 events each second. Did not stall the UI during stress testing. Real time probability charts showed a history of assault surges. The Live Attack Map showed the geographic origin of IP addresses.

6. DISCUSSION / ANALYSIS: This shows that Deep Learning can improve NIDS capabilities a lot. The CNN architecture performs better. The better performance means that the network traffic data has built-in patterns. The convolutional filters can decode the patterns. The convolutional filters work on the pixel layouts in photographs.

Alert fatigue, in Security Operations Centers (SOCs) is a problem. In my work I see that the live simulation engine fixes alert fatigue in Security Operations Centers (SOCs). AegisPulse reduces stress by turning text logs into charts such, as trend lines and heatmaps. AegisPulse does not make the operator read text. AegisPulse lets the operator see the speed of an attack and the color cues. The operator can quickly tell the intensity of an attack by looking at the speed and the color indicators.

7. RECOMMENDATIONS: To look at the context of the packets, over periods the future research should explore the transformer topologies (like BERT). Moving the backend to a distributed cloud architecture enables the scaling needed to handle enterprise traffic.

8. LIMITATIONS: I see that the report says the replayed datasets are used to approximate the real-time component. The report says the volatility and fragmentation, in the physical layer network traffic are not fully captured by the replayed datasets despite the accuracy. The report says the deep learning model is

computationally demanding and the deep learning models can be made for deployment, on edge devices with hardware resources.

9. CONCLUSION / SUMMARY: This research. Launched AegisPulse, a NIDS that mixes interactive visualization and hybrid deep learning. AegisPulse reaches, than 95 percent accuracy. AegisPulse provides a real time view of network activity. AegisPulse makes AI technology usable for cybersecurity work. The project shows that security tools need to be both easy, for users and work well in the algorithm.

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